

ANTONIO RAPUANO

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SUMMARY

I'm an engineer driven by the desire to turn rigorous theory into practical systems. I enjoy "full-stack" robotics — working across the entire development pipeline, from mathematical modeling and control design to embedded implementation and experimental validation. I'm dedicated to advancing the capabilities of autonomous systems through research and hands-on engineering.

EXPERIENCE

(Details and media at r-rapitano.com/about)

Research Engineer (Research Fellow)

Summer 2025 - Present

Sapienza University of Rome (uniroma1.it)

Highlights: Design and implementation of novel aerial interaction control strategies • Development of tactile sensing models using machine learning • Experimental validation on a hexarotor platform

Autopilot Control Systems Engineer

Fall 2024 - Summer 2025

Sapienza Flight Team (sasa-aerospace.it/flight-team)

Highlights: Design and implementation of path planning and hybrid control algorithms • Development of state estimation and filtering modules • Experimental testing on VTOL prototype

Race Timing Specialist and Team Lead

Fall 2018 - Present

ICRON (icron.it)

Highlights: Coordination of race logistics and technical setup • Supervision of start/finish line operations • Management of ranking and award procedures

EDUCATION

(Details and media at r-rapitano.com/about)

Ph.D. in Automatic Control, Bioengineering and Operations Research

Fall 2025 - Present

Sapienza University of Rome, Rome, Italy

M.Sc. in Control Engineering

Fall 2022 - Summer 2025

Sapienza University of Rome, Rome, Italy

Final grade: 110/110 with honors

Thesis: Nonlinear predictive control of the continuum and hybrid dynamics of a suspended deformable cable for aerial pick and place (Supervisor: Prof. Antonio Franchi)

Relevant coursework: Analysis, modeling and control of nonlinear, hybrid, multivariable, networked systems • Basic and advanced robotics • Estimation • Optimal control • Machine learning • Robust control

B.Sc. in Electronic Engineering for Automation and Telecommunications

Fall 2018 - Winter 2022

University of Sannio, Benevento, Italy

Final grade: 110/110 with honors

Thesis: Implementation on FPGA of an electronic circuit controlling a mechanical arm (Supervisor: Prof. Marco Pisco)

Relevant coursework: Analog and digital electronics • Analysis, modeling and control of linear systems • Digital design • Electronic measurements • Signal processing • Telecommunications

SKILLS

Software and tools: Arduino • Blender • Fusion 360 • LTSpice • Office • Quartus Prime • Simulink • Stateflow

Programming and formatting: C/C++ • Java • HTML/CSS • LaTeX • MATLAB • Python • Verilog

Hardware: 3D printing • Circuit board soldering • Measuring

Personal skills: Attention to detail • Autonomy and initiative • Critical thinking • Curiosity and continuous learning • Problem-solving attitude • Project planning • Technical writing • Time management

LANGUAGES

Italian, native

English, fluent

Cambridge English C1 Advanced - Score 207 (Grade A, CEFR Level C2)

Summer 2022